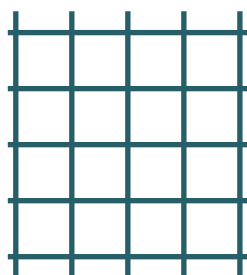


Retinal image size and aniseikonia

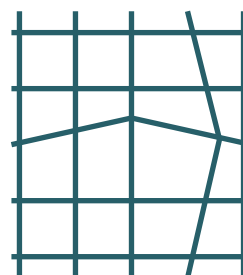
Part 1 | Equal prescriptions need not give equal image size

Spectacle magnification includes shape as well as power and vertex effects.

Uniform optical size



Retinal distortion



A global size factor may not correct a local deformation

Uniform optical magnification is distinguished from a field-dependent retinal size distortion.

The optical factors

For a spectacle model, magnification is the product of shape and power factors. Shape factor = $1/[1-(t/n)F1]$, with thickness t in metres, refractive index n and front-surface power $F1$. Equal back-vertex prescriptions do not force equal front curves or thickness.

Compare shape factors with fixed power factors

Holding the other variables constant isolates the front-surface contribution.

Original design comparison; $t=0.006$ m, $n=1.50$

Front power	Shape factor	Relative effect
+2 D	$1/(1-0.008)=1.008$	Lower magnification
+8 D	$1/(1-0.032)=1.033$	Higher magnification

Numerical interpretation

The second shape factor is about 2.5% larger relative to the first. This is a comparison within the specified spectacle model, not a prediction that every patient will perceive the same size difference: retinal organisation and binocular adaptation also contribute.

Retinal distortion can survive optical correction

Matching corneal-plane optics does not reverse a displaced photoreceptor mosaic.

Comparison

Feature	Optical mechanism	Retinal mechanism
Distribution	Often broadly uniform	Can vary across the field
Lens change	May alter size mismatch	May leave local distortion
Macular assessment	May be normal	Can show structural deformation

Use the persistence

If a size complaint persists after an appropriate contact-lens comparison and accompanies an epiretinal membrane, retinal distortion remains a plausible cause. Persistence does not prove that spectacles had no contribution; it shows why an exclusively optical explanation may be incomplete.

Measure horizontal and vertical effects

A single global percentage can conceal meridional or local differences.

Characterise the complaint

- Identify which eye perceives the larger image
↓
- Compare horizontal and vertical dimensions
↓
- Assess variation with tested visual field
↓
- Relate the pattern to macular structure

A useful distinction

A field-dependent discrepancy cannot necessarily be corrected everywhere by one uniform magnifying lens. A correction that matches one test region may worsen another. Record the tested region and method before comparing numerical percentages across visits.

Acuity and binocular comfort are different outcomes

Improved letter recognition does not guarantee restoration of perceived size matching.

Outcome

Feature	Measurement	Possible dissociation
Monocular acuity	Letters recognised	Can improve despite distortion
Metamorphopsia	Shape deformation	May remain
Aniseikonia	Interocular perceived size	May remain field-dependent

Counselling from mechanism

Explain which component an optical correction is intended to change. A technically adequate refraction can coexist with a retinally mediated binocular symptom; documenting both avoids labelling the residual complaint as evidence that the prescription must be wrong.